

1c. Query Data in a Lakehouse SQL Analytics Endpoint

The SQL Analytics Endpoint is a read-only SQL editor that provides an interface to query lakehouse data with T-SQL.

For this lab, we'll just explore the data. Feel free to write other queries if you like. The subsequent labs do not depend on this lab.

Explore the environment

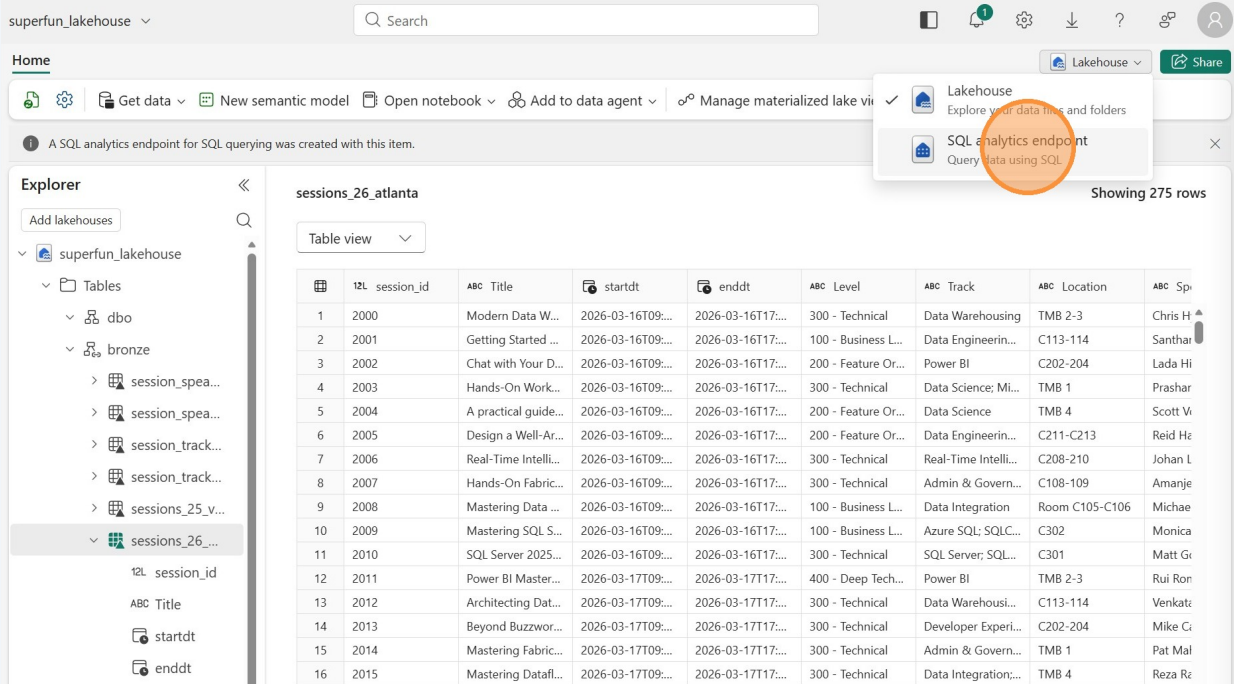
1 Start in your lakehouse.

The screenshot displays the Lakehouse SQL Analytics Endpoint interface. On the left, the Explorer pane shows a tree structure with 'superfun_lakehouse' expanded, revealing 'Tables' and 'Bronze' folders. The 'Tables' folder is expanded, showing 'session_id', 'ABC Title', 'startdt', and 'enddt'. The 'Bronze' folder is also expanded, showing 'session_id', 'ABC Title', 'startdt', and 'enddt'. The main area displays a table view of 'sessions_26_atlanta' with 275 rows. The table has columns: session_id, ABC Title, startdt, enddt, Level, Track, Location, and Sp. An orange circle highlights the 'Lakehouse' dropdown menu in the top right corner of the interface.

	session_id	ABC Title	startdt	enddt	Level	Track	Location	Sp
1	2000	Modern Data W...	2026-03-16T09:...	2026-03-16T17:...	300 - Technical	Data Warehousing	TMB 2-3	Chris H
2	2001	Getting Started ...	2026-03-16T09:...	2026-03-16T17:...	100 - Business L...	Data Engineerin...	C113-114	Santha
3	2002	Chat with Your D...	2026-03-16T09:...	2026-03-16T17:...	200 - Feature Or...	Power BI	C202-204	Lada Hi
4	2003	Hands-On Work...	2026-03-16T09:...	2026-03-16T17:...	300 - Technical	Data Science; Mi...	TMB 1	Prashar
5	2004	A practical guide...	2026-03-16T09:...	2026-03-16T17:...	200 - Feature Or...	Data Science	TMB 4	Scott V
6	2005	Design a Well-Ar...	2026-03-16T09:...	2026-03-16T17:...	200 - Feature Or...	Data Engineerin...	C211-C213	Reid He
7	2006	Real-Time Intelli...	2026-03-16T09:...	2026-03-16T17:...	300 - Technical	Real-Time Intelli...	C208-210	Johan L
8	2007	Hands-On Fabric...	2026-03-16T09:...	2026-03-16T17:...	300 - Technical	Admin & Govern...	C108-109	Amanje
9	2008	Mastering Data ...	2026-03-16T09:...	2026-03-16T17:...	100 - Business L...	Data Integration	Room C105-C106	Michae
10	2009	Mastering SQL S...	2026-03-16T09:...	2026-03-16T17:...	100 - Business L...	Azure SQL; SQLC...	C302	Monica
11	2010	SQL Server 2025...	2026-03-16T09:...	2026-03-16T17:...	300 - Technical	SQL Server; SQL...	C301	Matt Gr
12	2011	Power BI Master...	2026-03-17T09:...	2026-03-17T17:...	400 - Deep Tech...	Power BI	TMB 2-3	Rui Ron
13	2012	Architecting Dat...	2026-03-17T09:...	2026-03-17T17:...	300 - Technical	Data Warehousi...	C113-114	Venkatz
14	2013	Beyond Buzzwor...	2026-03-17T09:...	2026-03-17T17:...	300 - Technical	Developer Experi...	C202-204	Mike G
15	2014	Mastering Fabric...	2026-03-17T09:...	2026-03-17T17:...	300 - Technical	Admin & Govern...	TMB 1	Pat Mal
16	2015	Mastering Datafl...	2026-03-17T09:...	2026-03-17T17:...	300 - Technical	Data Integration;...	TMB 4	Reza R

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Click the "Lakehouse" dropdown in the top-right of the screen to see your options. Click "SQL analytics endpoint" to go to the SQL analytics endpoint from the lakehouse.

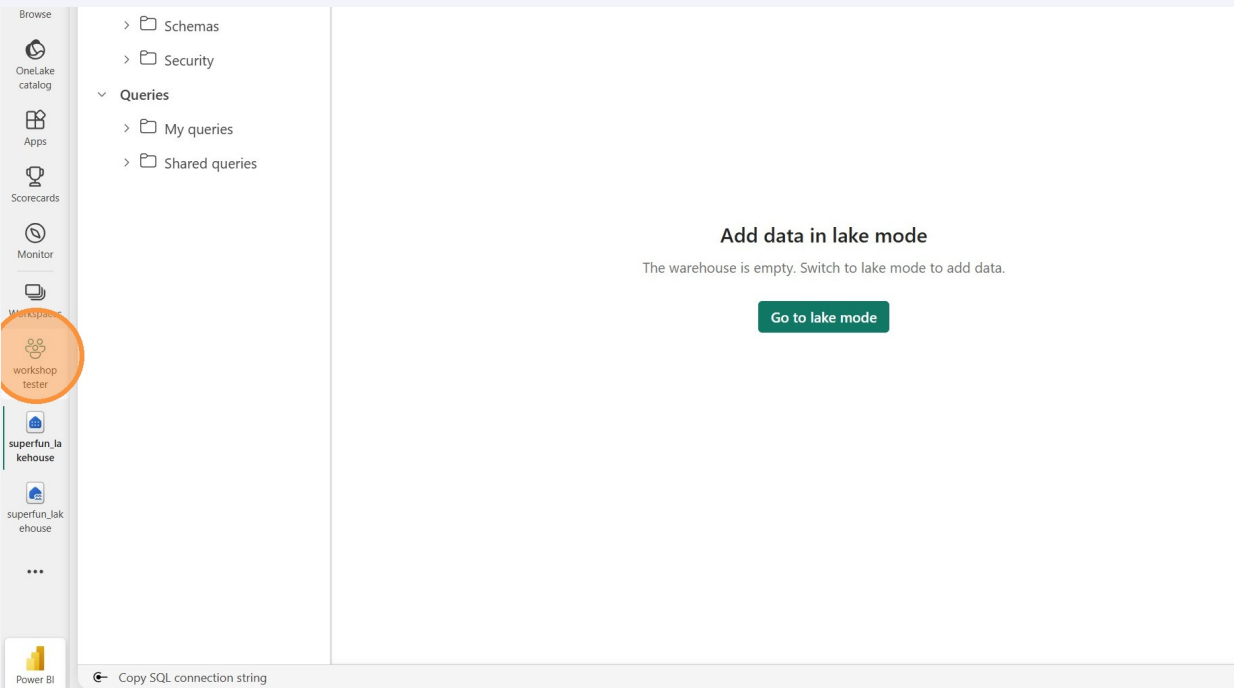


The screenshot shows the Lakehouse interface with the "Lakehouse" dropdown menu open. The menu options are "Lakehouse" (Explore your data files and folders) and "SQL analytics endpoint" (Query data using SQL). The "SQL analytics endpoint" option is highlighted with an orange circle. The main view displays a table named "sessions_26_atlanta" with 275 rows. The table columns are: session_id, ABC Title, startdt, enddt, ABC Level, ABC Track, ABC Location, and ABC Sp.

	session_id	ABC Title	startdt	enddt	ABC Level	ABC Track	ABC Location	ABC Sp
1	2000	Modern Data W...	2026-03-16T09:...	2026-03-16T17:...	300 - Technical	Data Warehousing	TMB 2-3	Chris H
2	2001	Getting Started ...	2026-03-16T09:...	2026-03-16T17:...	100 - Business L...	Data Engineerin...	C113-114	Santhar
3	2002	Chat with Your D...	2026-03-16T09:...	2026-03-16T17:...	200 - Feature Or...	Power BI	C202-204	Lada Hi
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5	2004	A practical guide...	2026-03-16T09:...	2026-03-16T17:...	200 - Feature Or...	Data Science	TMB 4	Scott V
6	2005	Design a Well-Ar...	2026-03-16T09:...	2026-03-16T17:...	200 - Feature Or...	Data Engineerin...	C211-C213	Reid H
7	2006	Real-Time Intelli...	2026-03-16T09:...	2026-03-16T17:...	300 - Technical	Real-Time Intelli...	C208-210	Johan L
8	2007	Hands-On Fabric...	2026-03-16T09:...	2026-03-16T17:...	300 - Technical	Admin & Govern...	C108-109	Amanje
9	2008	Mastering Data ...	2026-03-16T09:...	2026-03-16T17:...	100 - Business L...	Data Integration	Room C105-C106	Michae
10	2009	Mastering SQL S...	2026-03-16T09:...	2026-03-16T17:...	100 - Business L...	Azure SQL; SQLC...	C302	Monica
11	2010	SQL Server 2025...	2026-03-16T09:...	2026-03-16T17:...	300 - Technical	SQL Server; SQL...	C301	Matt Gr
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13	2012	Architecting Dat...	2026-03-17T09:...	2026-03-17T17:...	300 - Technical	Data Warehousi...	C113-114	Venkatz
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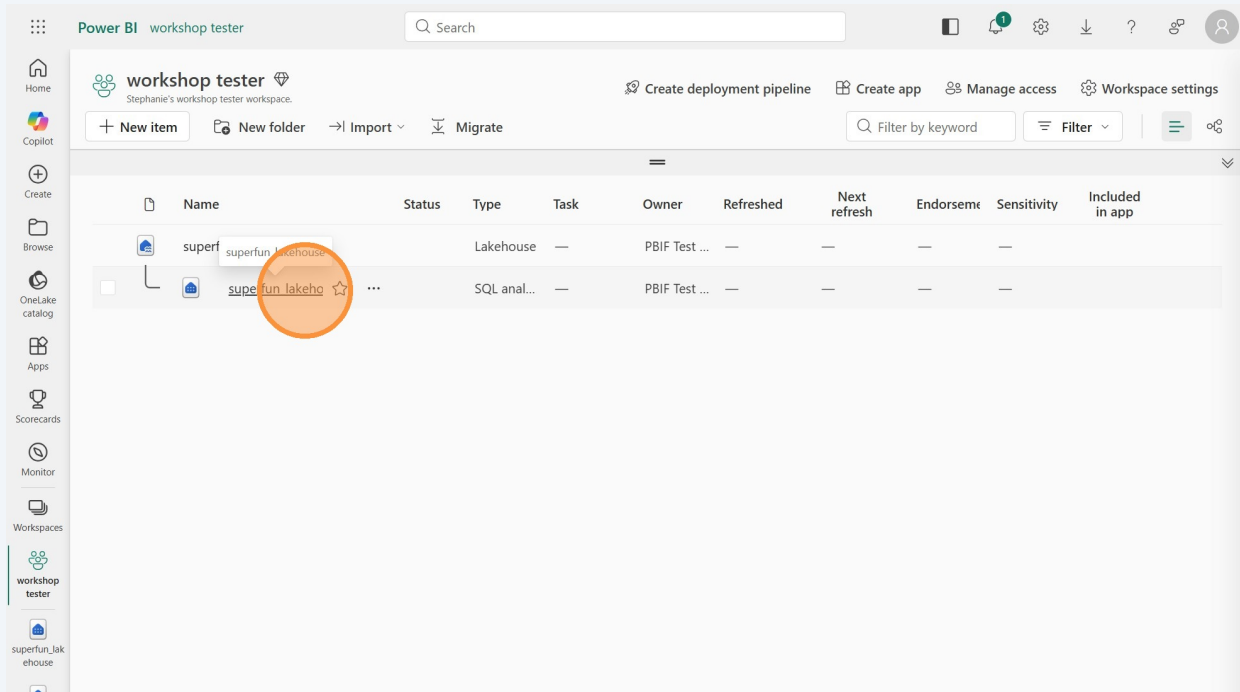
Alternatively, go back to the workspace. If you're not already in the lakehouse, this is a good way to get to the SQL analytics endpoint.



The screenshot shows the Lakehouse interface in workspace view. The left sidebar contains a list of items: Schemas, Security, Queries, My queries, Shared queries, workshop tester (highlighted with an orange circle), superfun_la kehouse, superfun_lak ehouse, and Power BI. The main view displays a message: "Add data in lake mode" and "The warehouse is empty. Switch to lake mode to add data." with a "Go to lake mode" button.

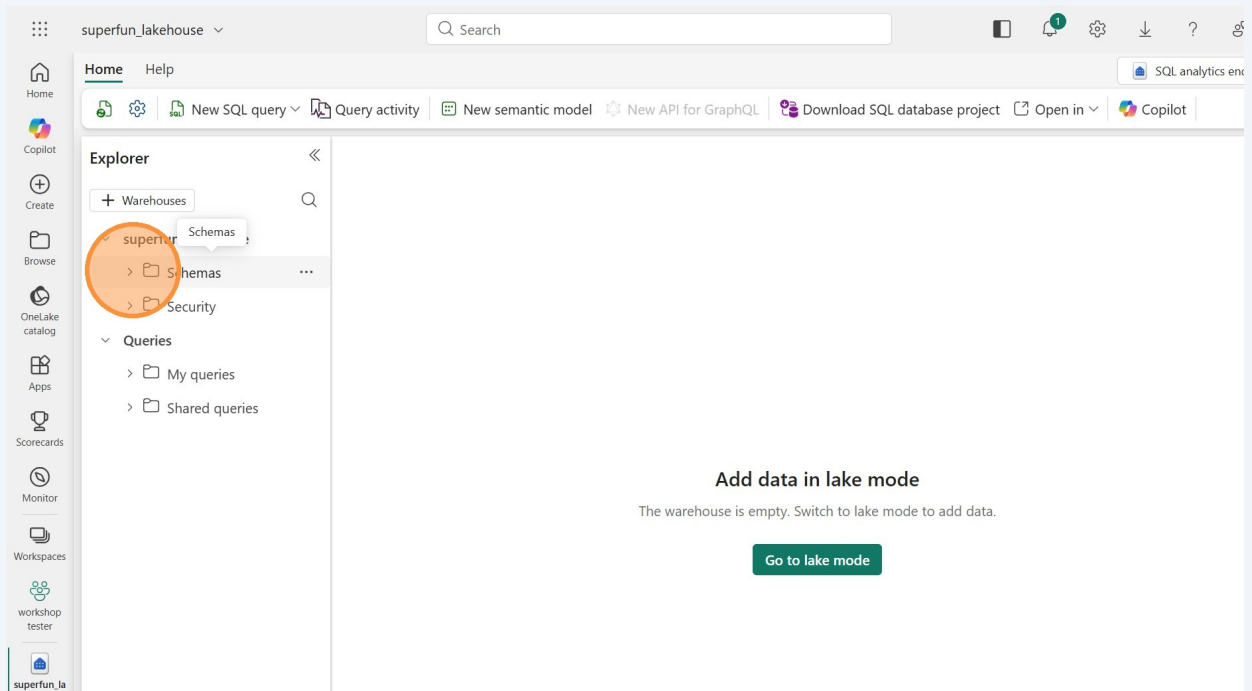
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Click the **SQL analytics endpoint** for your lakehouse in the workspace.



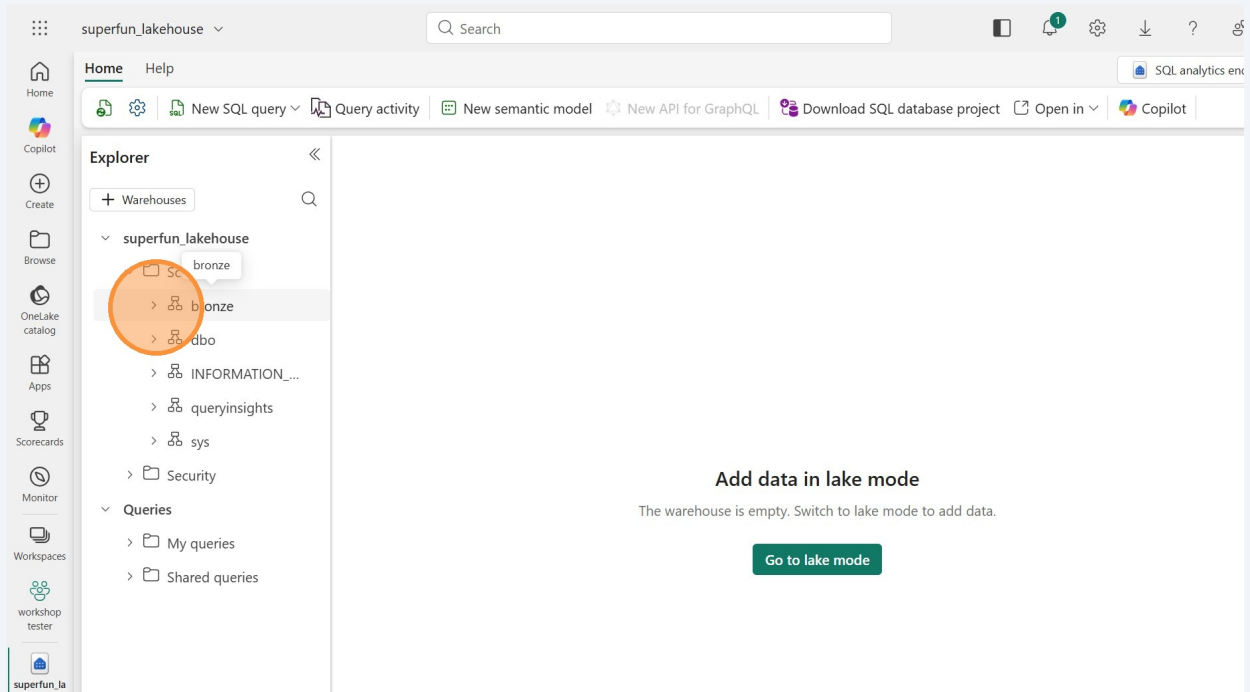
5

In the Explorer section, notice that it looks very similar to what we would see in SQL Server Management Studio. Expand "Schemas."



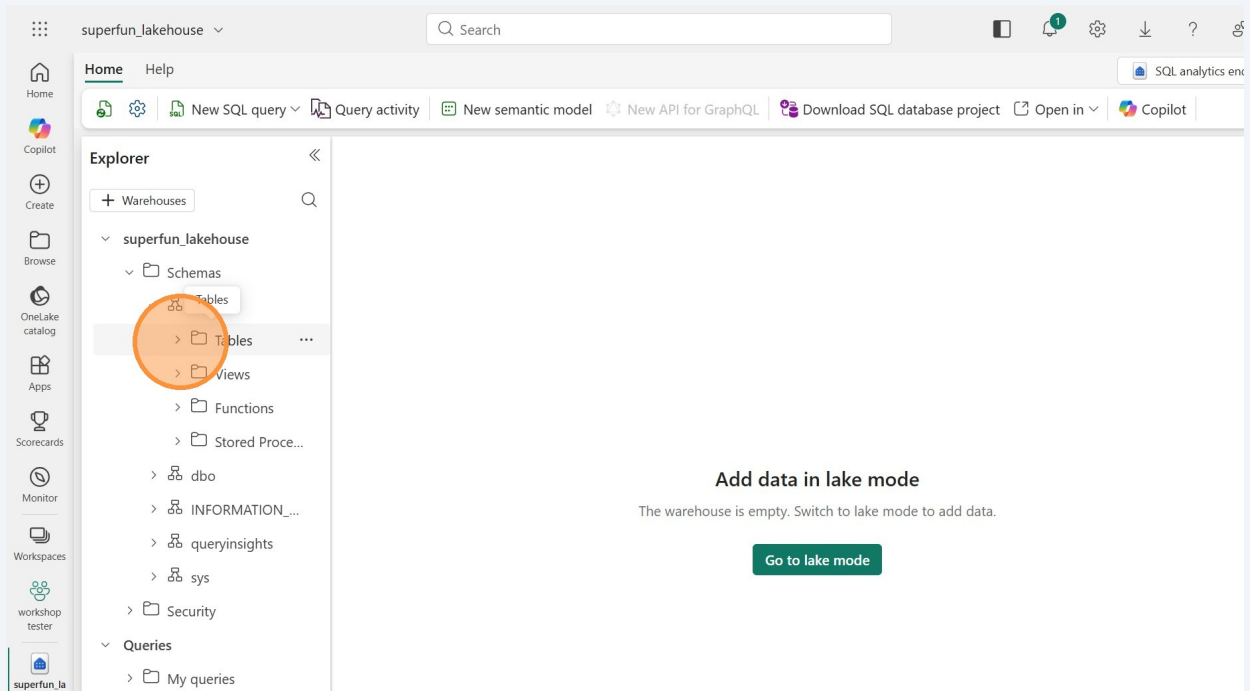
6

Expand the bronze schema to see the shortcutted tables. (is shortcutted a word?)



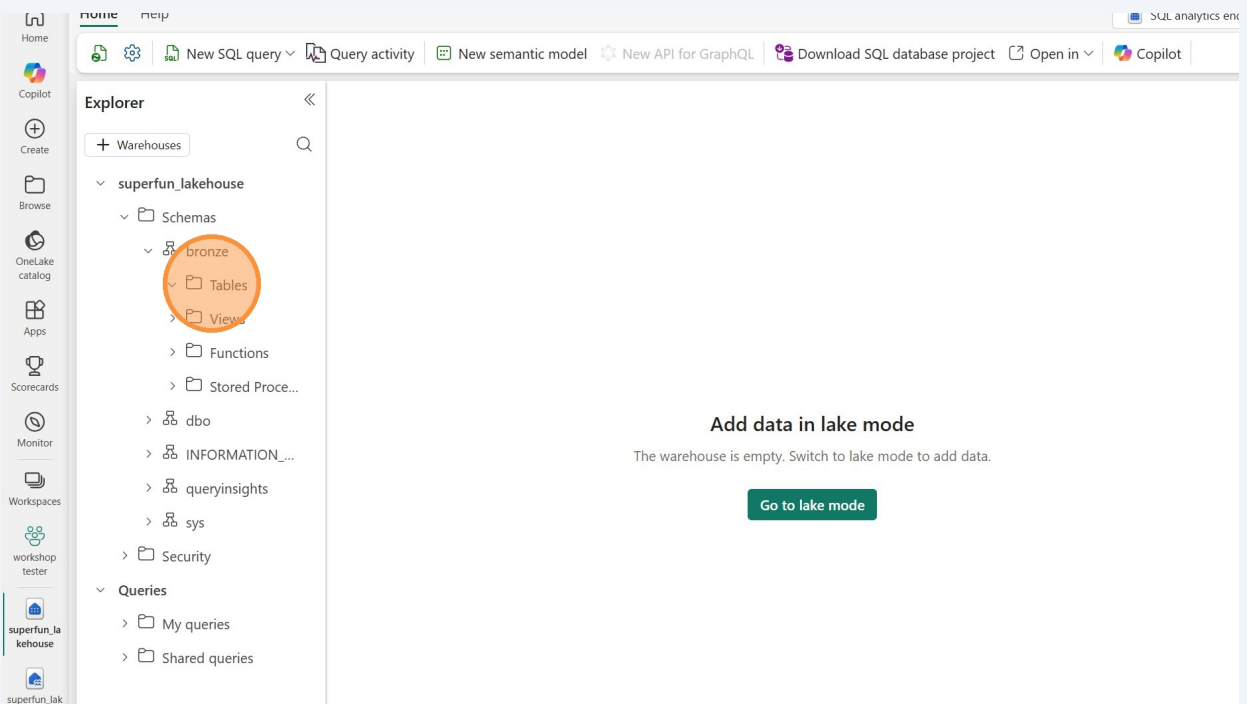
7

Under the "bronze" schema, we have folders for Tables, Views, Functions, and Stored Procedures. Expand the "Tables" folder.



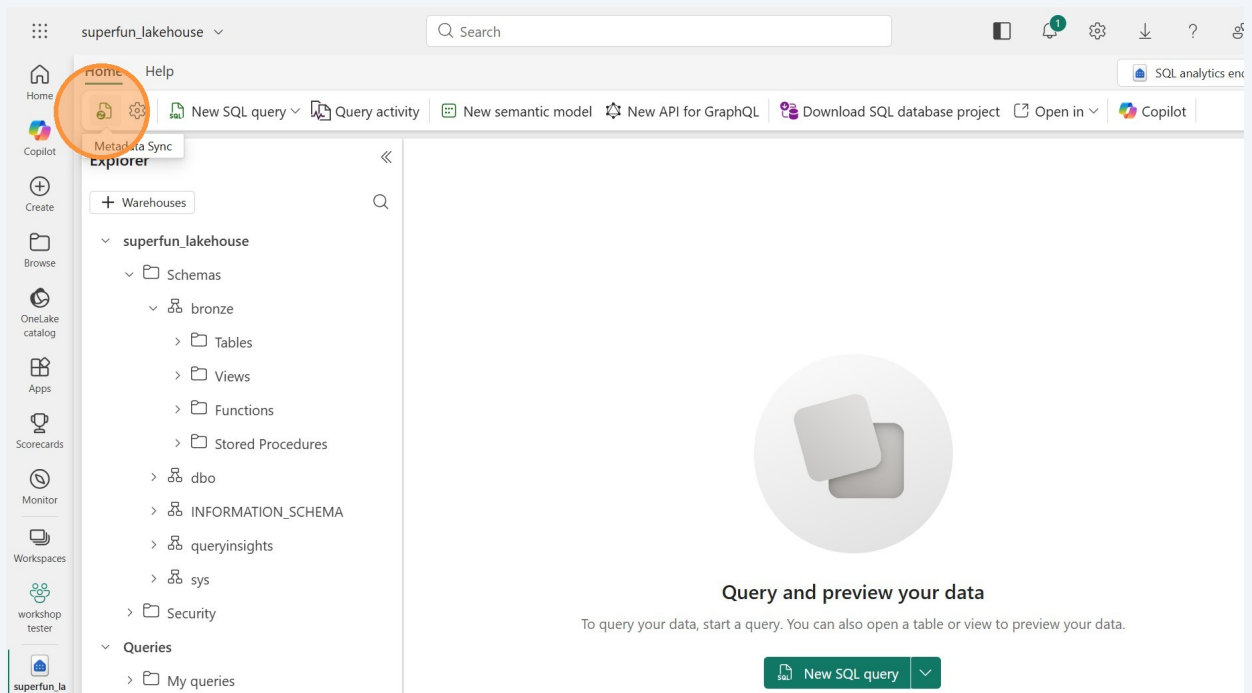
8

It's possible that we can't see the tables yet, since we just created them. If that you don't see any, we'll just need to refresh the analytics endpoint view.

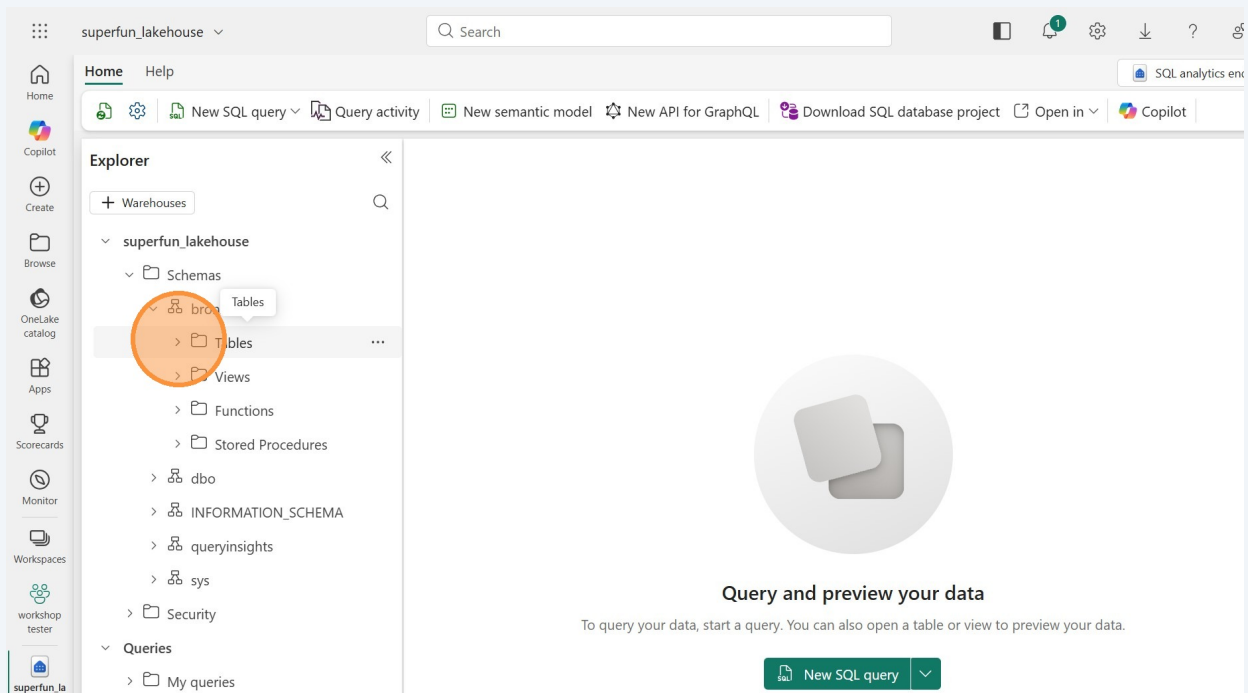


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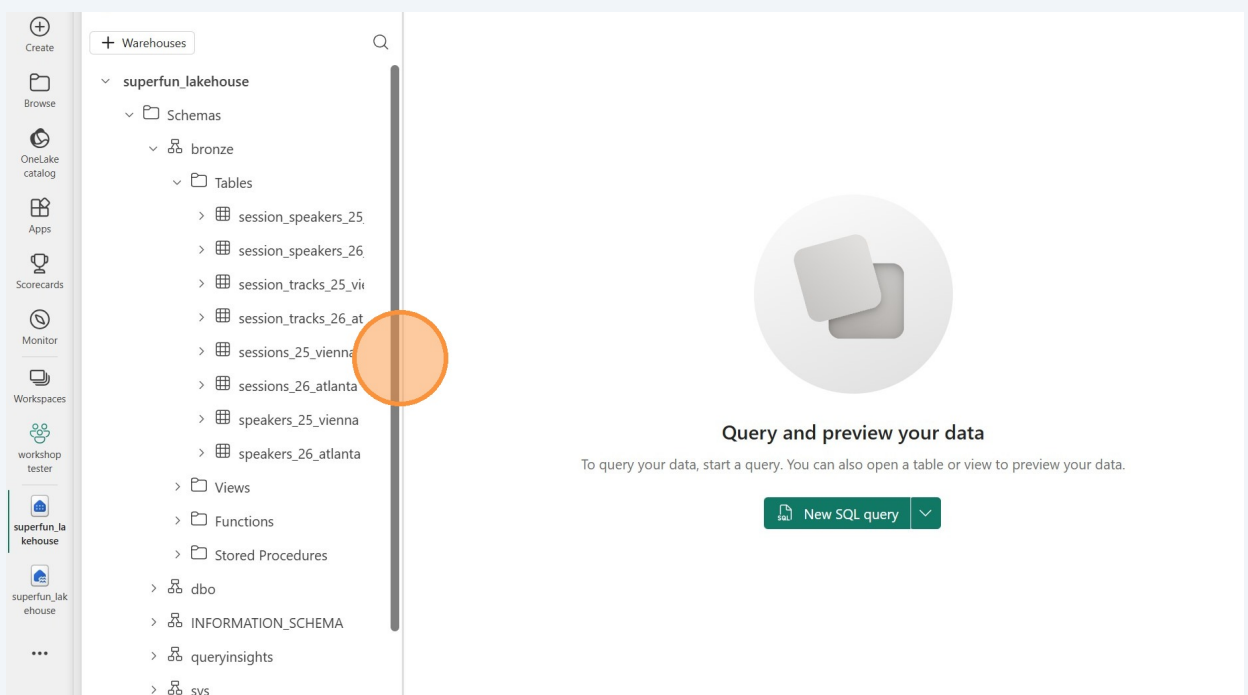
Click this little green refresh button under the Home tab.



10 Now expand the Tables folder again.



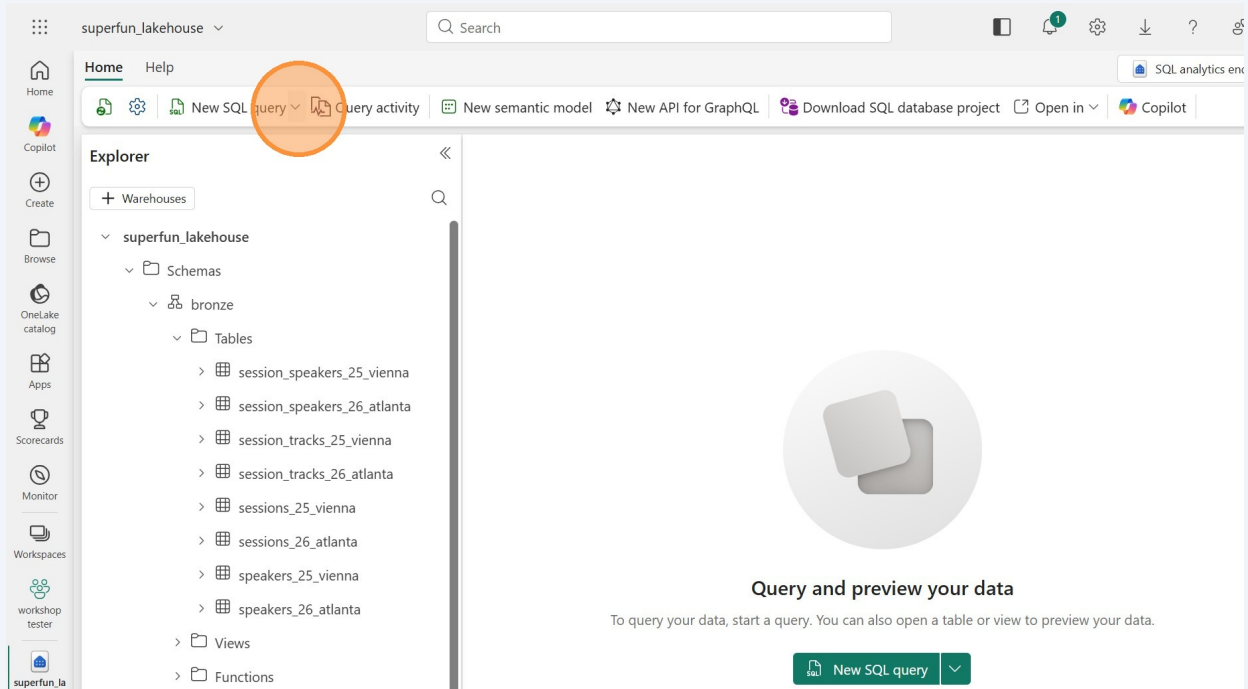
11 We see our eight tables. Since it's hard to read the full table names, you might want to click and drag the horizontal bar to the right to see the full table names.



Write a SQL query

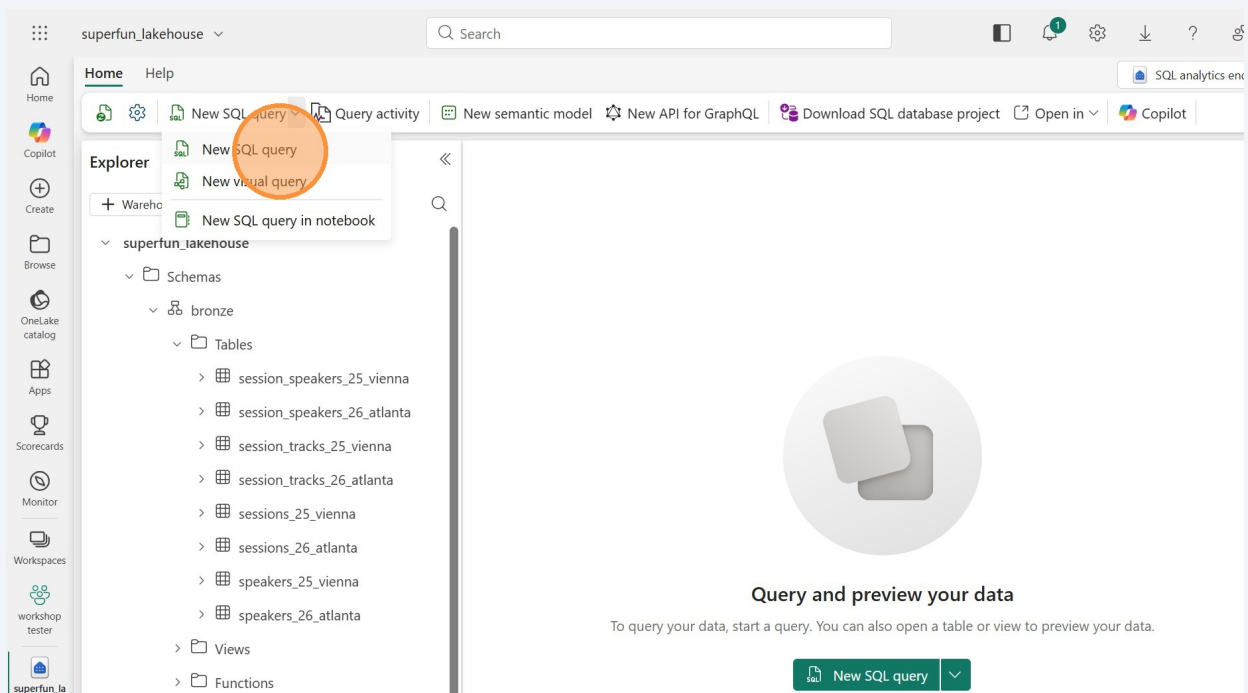
12

Now let's query the data. You can either click the big green "New SQL query" button in the middle of the canvas, or click the "SQL" dropdown in the ribbon under the Home tab.

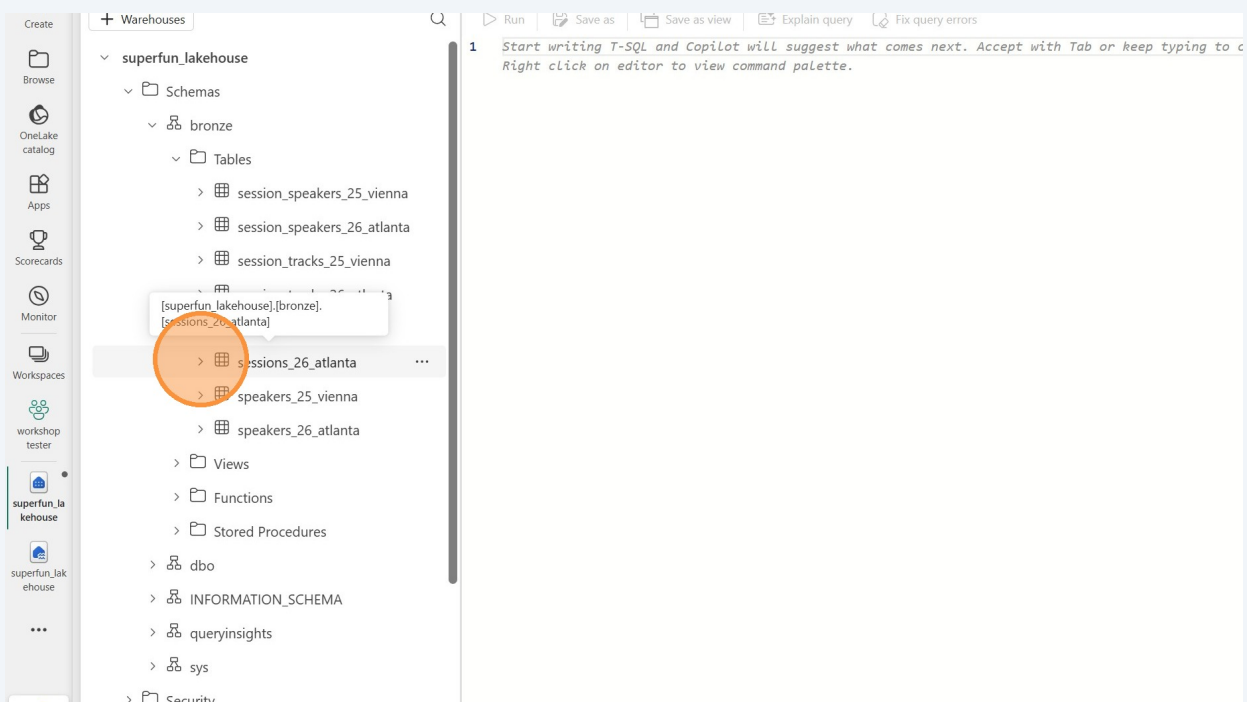


13

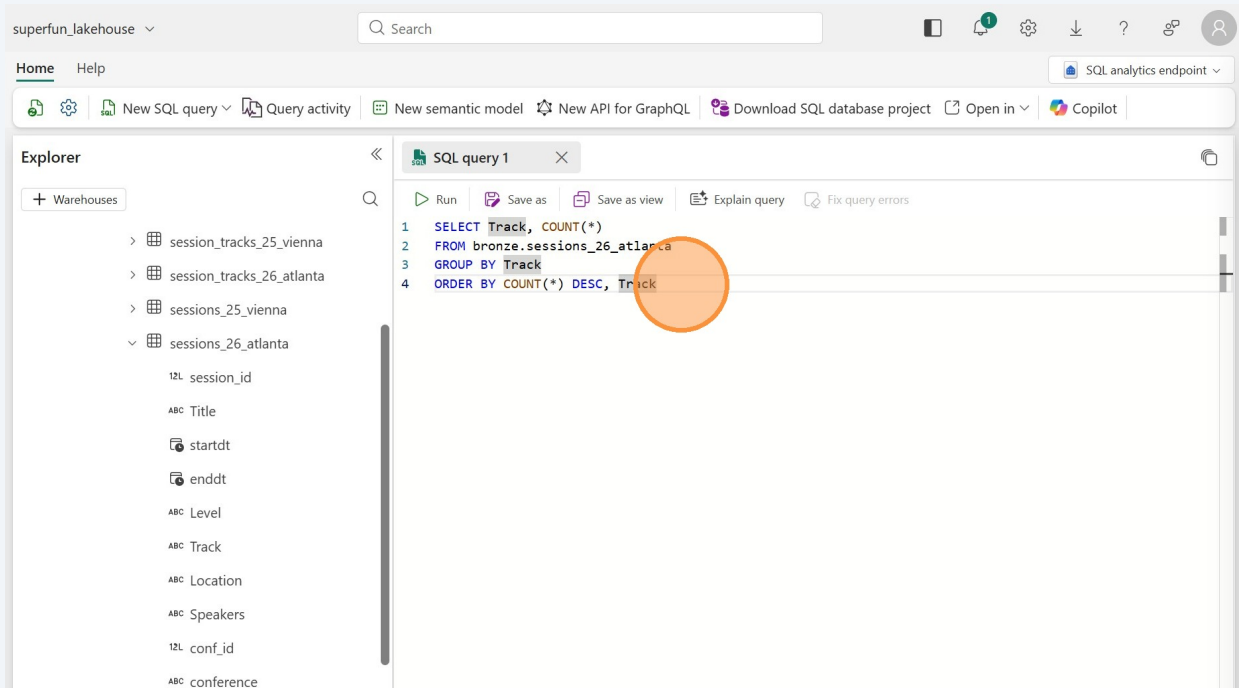
Click "New SQL query". We'll take a look at the visual queries in the next section. Note that you can also do a query in a notebook, but we aren't covering that today.



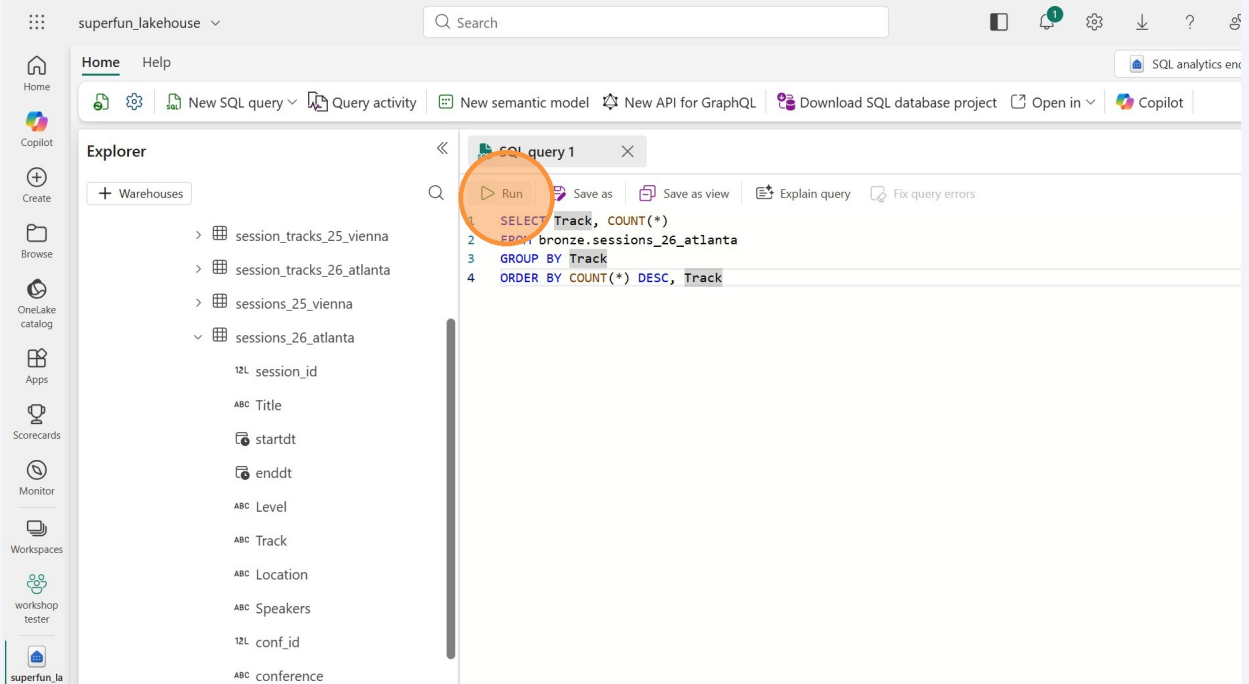
14 Before we write our query, let's expand the table to see the column names.



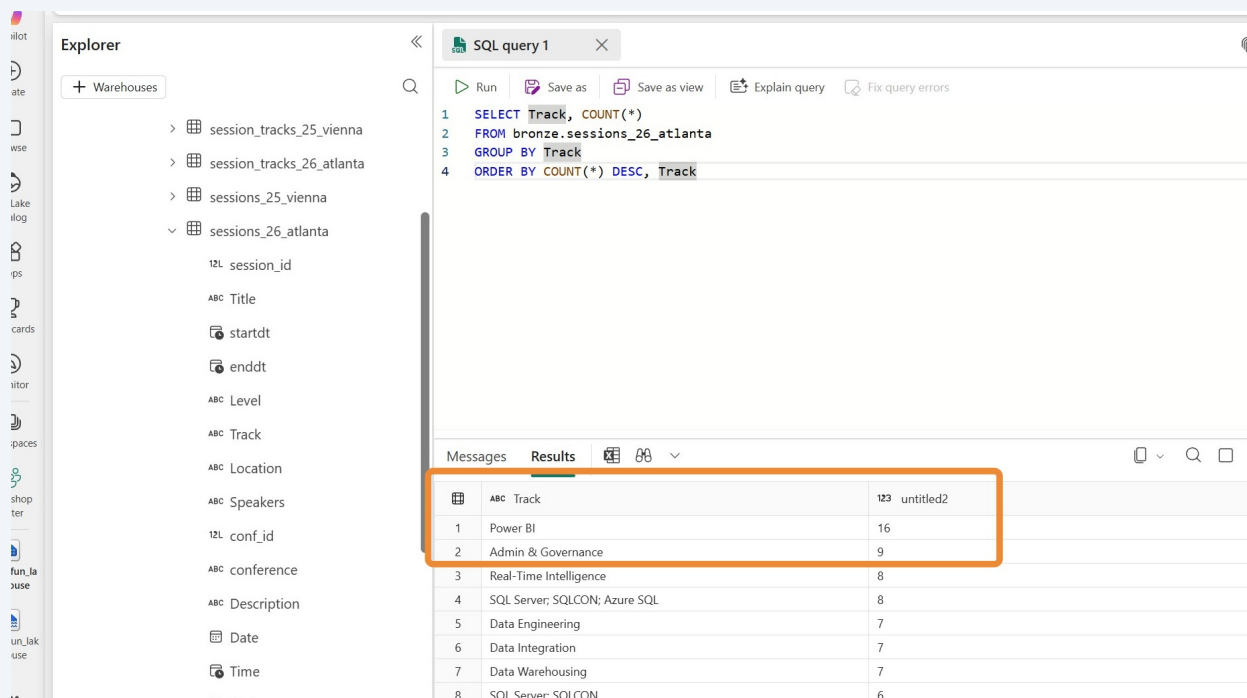
15 In the query editor (with a tab called SQL query 1), type of paste in the following sample query:



16 Click "Run"

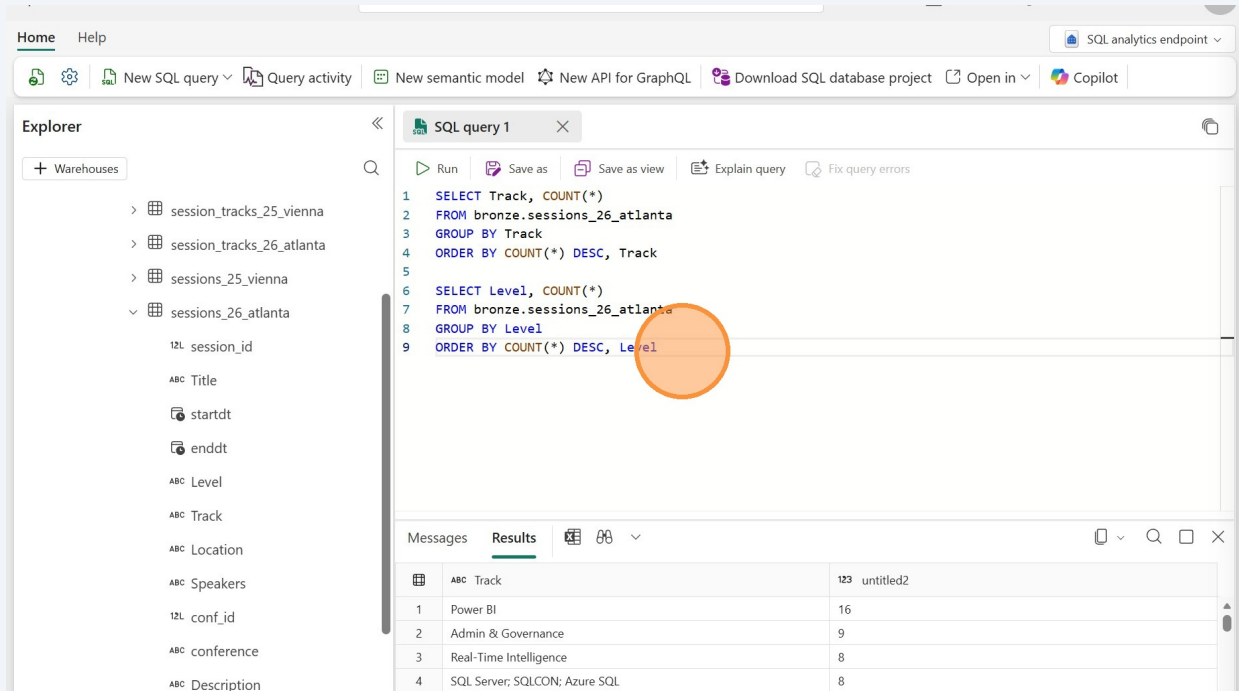


17 In the output we can see that the Power BI track has the most sessions, followed by Admin & Governance. Real-Time Intelligence takes third place.



18

Let's do another query to see sessions by level. Copy the first query and paste it below the first query. Change "Track" to "Level" on the SELECT, GROUP BY, and ORDER BY lines. Or, just paste in the following:



The screenshot shows the SQL query editor with two queries. The first query is for sessions by track, and the second is for sessions by level. An orange circle highlights the second query.

```

1 SELECT Track, COUNT(*)
2 FROM bronze.sessions_26_atlanta
3 GROUP BY Track
4 ORDER BY COUNT(*) DESC, Track
5
6 SELECT Level, COUNT(*)
7 FROM bronze.sessions_26_atlanta
8 GROUP BY Level
9 ORDER BY COUNT(*) DESC, Level

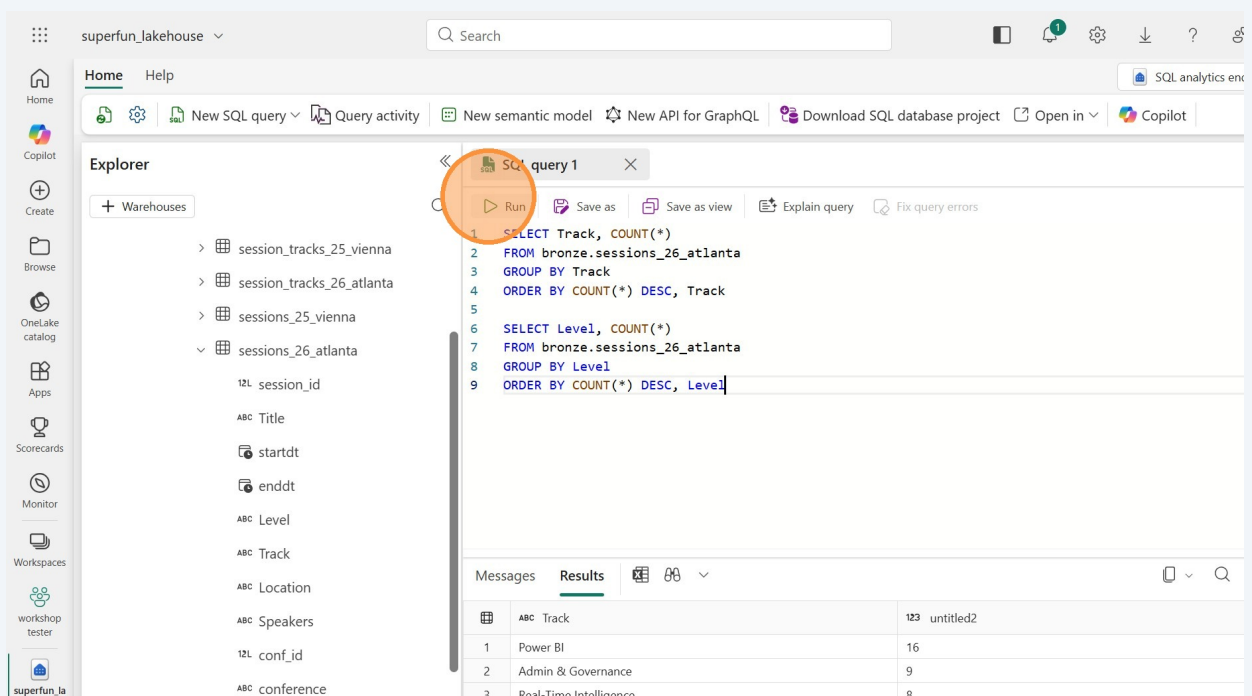
```

The results table shows the following data:

ABC Track	123 untitled2
1 Power BI	16
2 Admin & Governance	9
3 Real-Time Intelligence	8
4 SQL Server; SQLCON: Azure SQL	8

19

Click "Run".



The screenshot shows the SQL query editor with the 'Run' button highlighted with an orange circle. The query is the same as in the previous screenshot.

```

1 SELECT Track, COUNT(*)
2 FROM bronze.sessions_26_atlanta
3 GROUP BY Track
4 ORDER BY COUNT(*) DESC, Track
5
6 SELECT Level, COUNT(*)
7 FROM bronze.sessions_26_atlanta
8 GROUP BY Level
9 ORDER BY COUNT(*) DESC, Level

```

The results table shows the following data:

ABC Track	123 untitled2
1 Power BI	16
2 Admin & Governance	9
3 Real-Time Intelligence	8

20

Now that we have two queries, we will get two results below. We have a dropdown to go back and forth between the two results. Right now we see the results of the first query. Click the dropdown.

The screenshot shows a SQL IDE interface. On the left is a file explorer with a tree view containing folders like 'session_tracks_25_vienna', 'session_tracks_26_atlanta', 'sessions_25_vienna', and 'sessions_26_atlanta'. The main editor displays two SQL queries. The first query is selected, and its results are shown in a table below. The table has two columns: 'Track' and 'Count'. The results are as follows:

Track	Count
Power BI	16
Admin & Governance	9
Real-Time Intelligence	8
SQL Server; SQLCON; Azure SQL	8
Data Engineering	7
Data Integration	7
Data Warehousing	7
SQL Server; SQLCON	6
Azure SQL; SQLCON; SQL Server	5
Data Engineering; Data Integration	5
Data Science	5

A dropdown arrow in the top right corner of the results pane is highlighted with an orange circle.

21

Click "Result: 2" to see the results of the Level query.

The screenshot shows the same SQL IDE interface as before, but now the dropdown menu is open, and 'Result: 2' is selected. The results of the second query are displayed in the table below. The table has two columns: 'Level' and 'Count'. The results are as follows:

Level	Count
1	16
2	9
3	8
4	8
5	7
6	7
7	7
8	6
9	5
10	5
11	5

The dropdown menu is highlighted with an orange circle, showing 'Result: 1' and 'Result: 2'.

Create a Visual Query

22

Another way to explore the data is with a visual query. This is really helpful for people who aren't comfortable writing SQL queries because it gives them a visual editor, like Power Query. Click the "SQL" button in the ribbon and then "New visual query."

The screenshot shows the Microsoft Fabric interface with the 'superfun_lakehouse' selected. In the 'Home' ribbon, the 'New SQL query' dropdown menu is open, and 'New visual query' is highlighted with an orange circle. The main workspace displays 'SQL query 1' with the following SQL code:

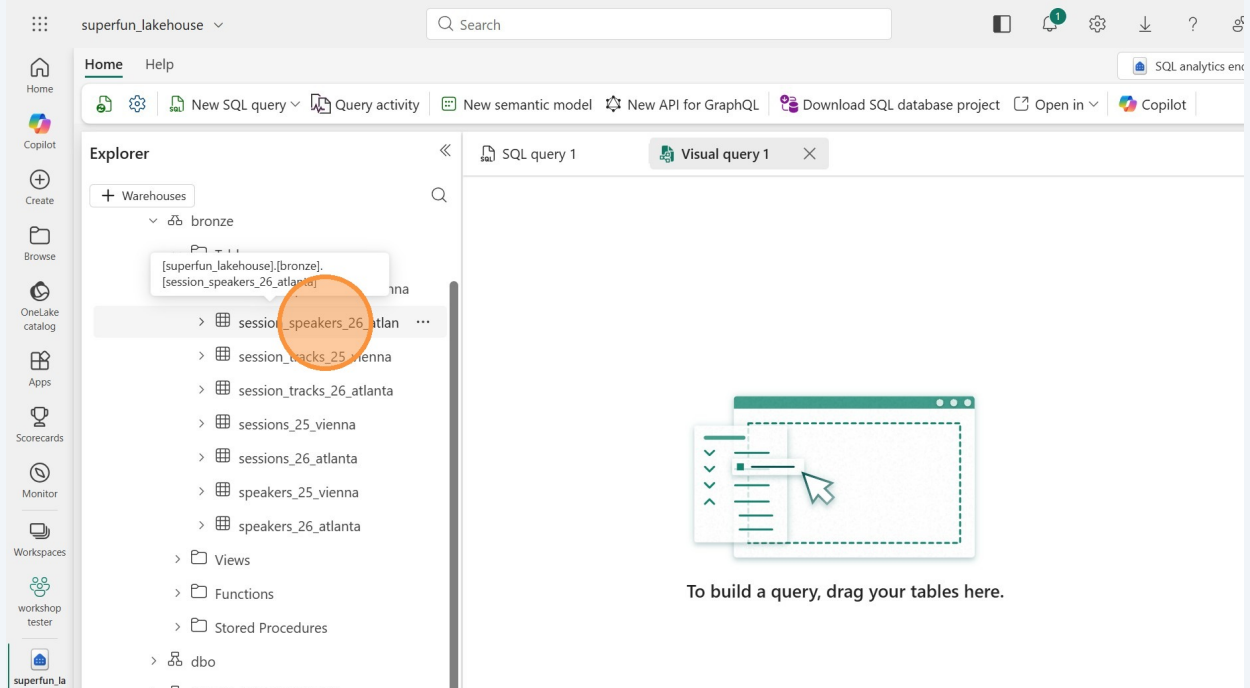
```
1 SELECT Track, COUNT(*)
2 FROM bronze.sessions_26_atlanta
3 GROUP BY Track
4 ORDER BY COUNT(*) DESC, Track
5
6 SELECT Level, COUNT(*)
7 FROM bronze.sessions_26_atlanta
8 GROUP BY Level
9 ORDER BY COUNT(*) DESC, Level
```

Below the query editor, the 'Results' tab is active, showing a table with 3 rows and 2 columns. The table is titled '123 untitled2'.

ABC Level	
1 300 - Technical	114
2 200 - Feature Oriented	108
3 100 - Business Level	37

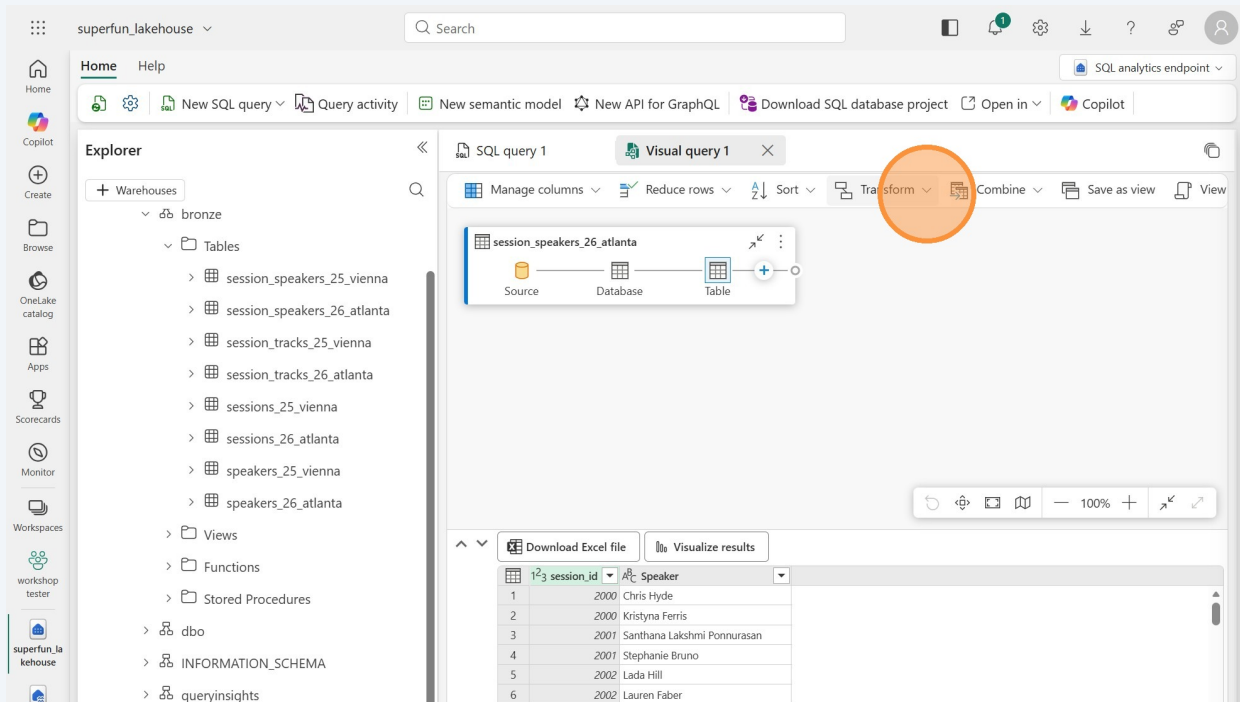
23

Click "session_speakers_26_atlanta" and drag it onto the canvas (on the right side of the window).

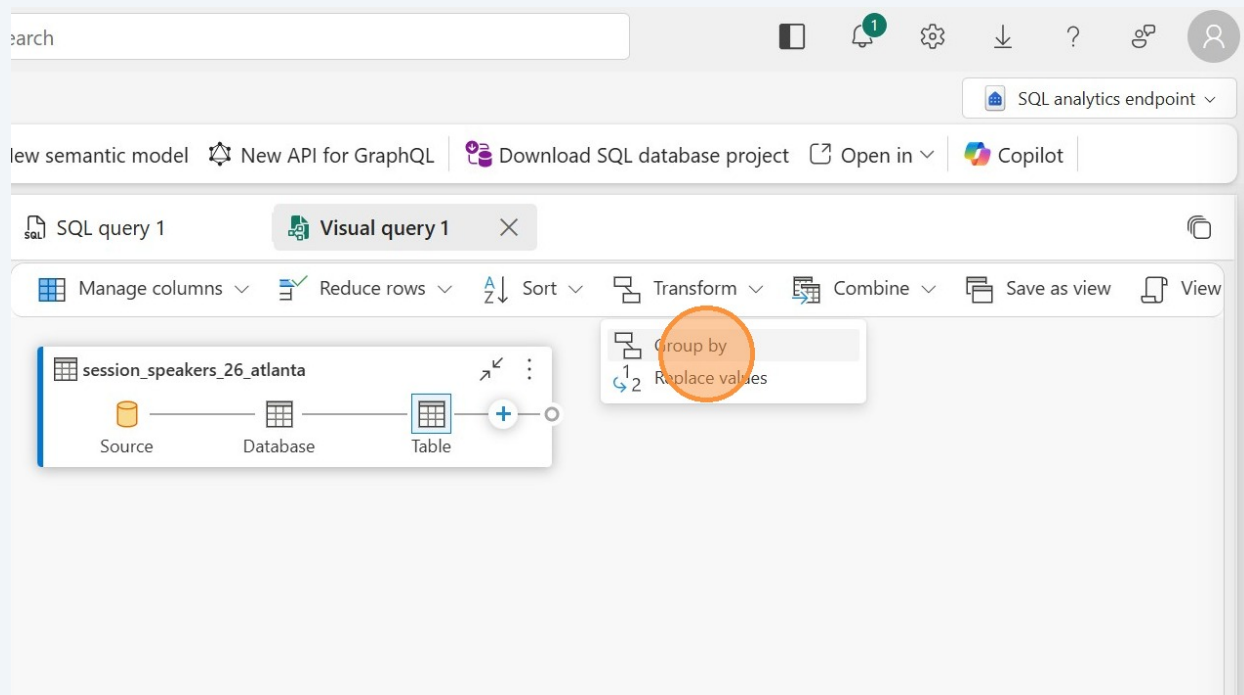


24

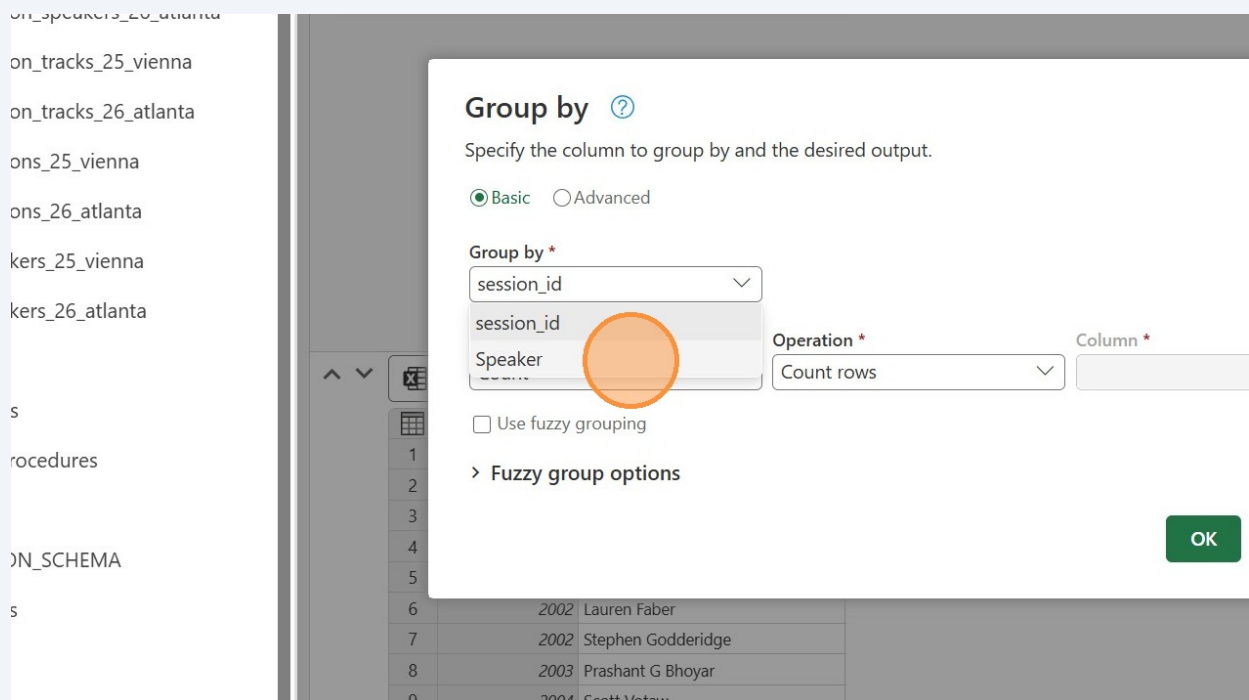
Let's group by speaker name to see which speakers are presenting the most sessions. Click the Transform button.



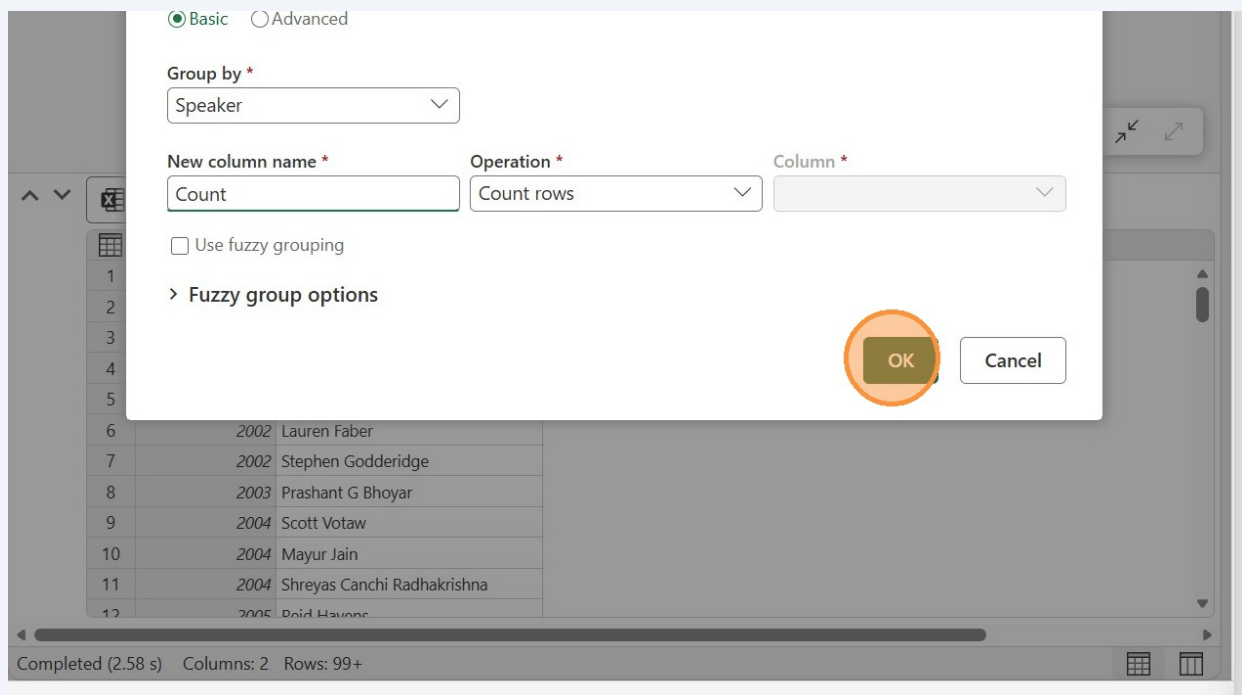
25 Click "Group by"



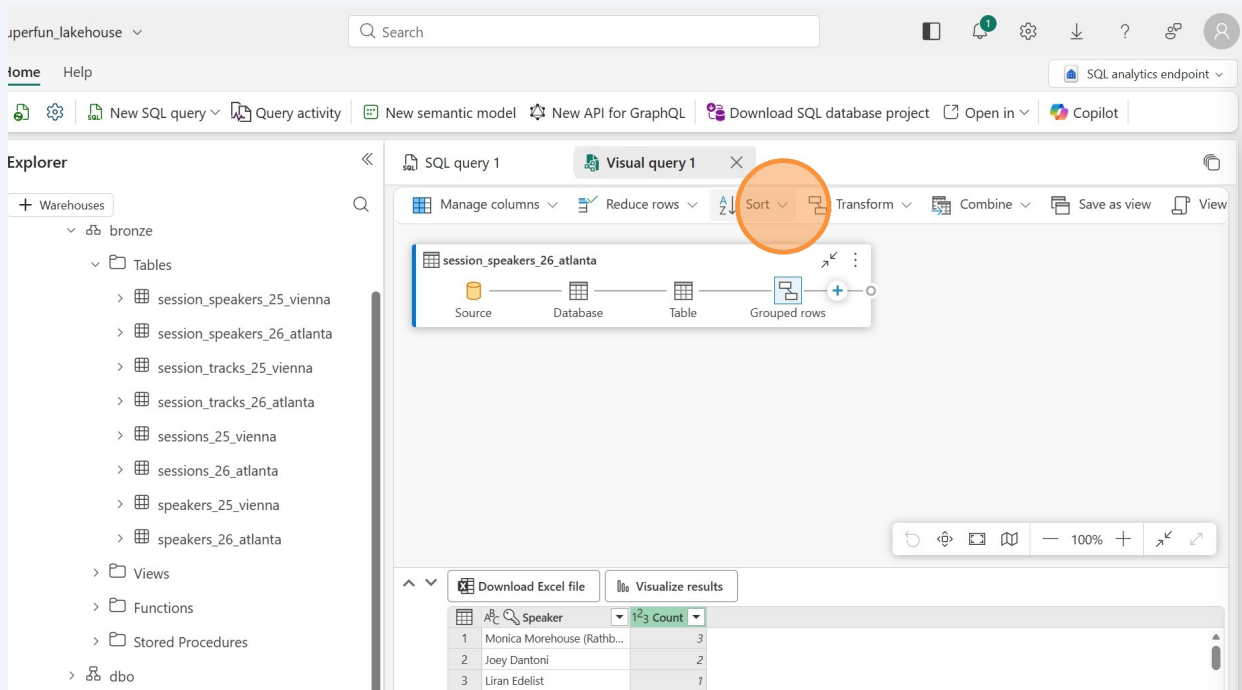
26 Click the "Group by" dropdown and choose "speaker_name."



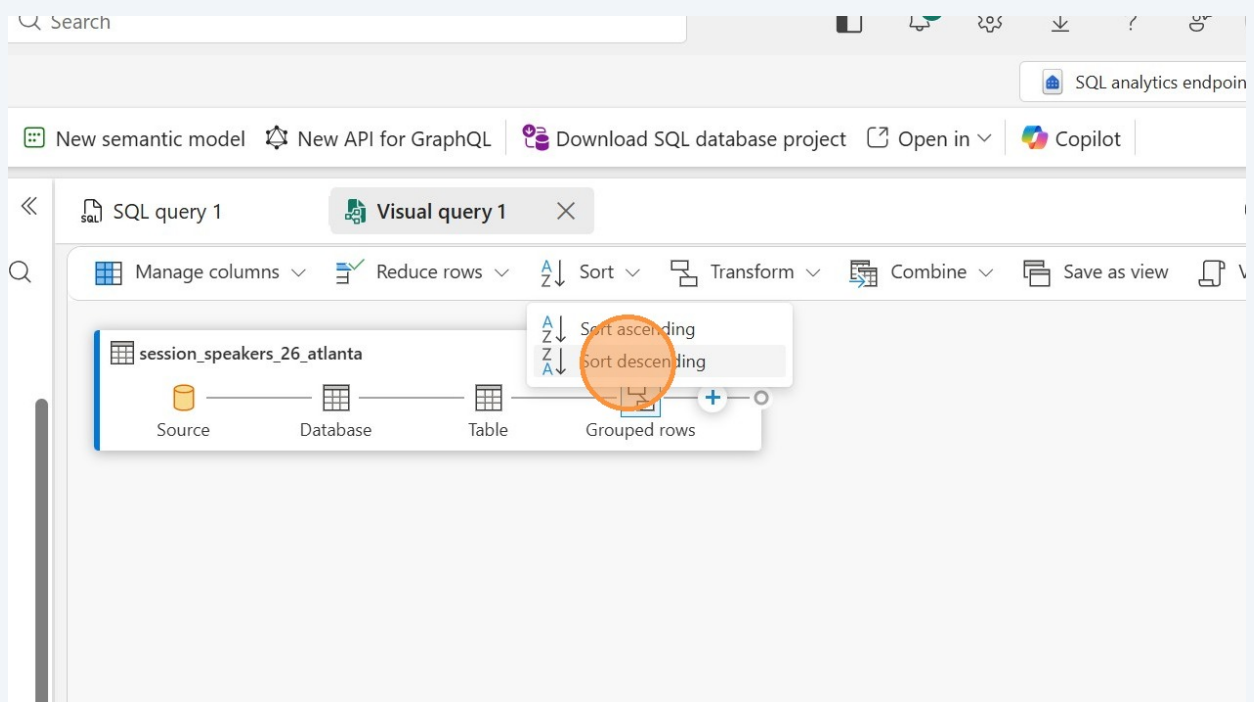
27 Click "OK"



28 We have the results, but it's sorted by speaker. Click the "Sort" button.



29 Click "Sort descending"



30 We can see that Bob Ward is presenting four sessions.

